

Phu Vinh Nguyen

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SKILLS

Languages: JavaScript, HTML, CSS, Bash/Shell, Python, SQL

Frameworks and Libraries: React, CustomTkinter, Express, Electron

Technical Tools: Git, Jira, HPC, User-level LIMS, Docker

Flexible to learn new languages / framework/ system, eager to learn more

WORK EXPERIENCE

Automation Engineer | Developer, **Sanofi Pasteur**

Jun 2025 – Present

Automation & Data Engineering Team

- Developed a form request desktop app using React and Electron to standardize researcher form submissions and automate internal ticket creation via SMTP email integration.

Automation Engineer | Developer, **Foundation Medicine Inc**

Nov 2024 – Apr 2025

Assay Development, Automation Engineering Team

- Developed and launched a Python-GUI app using CustomTkinter to provide a clean interface for backend automation scripts to enable non-coders to perform tasks independently.
- Built a React troubleshooting app for Hamilton devices that displays the metadata from specific objects to help cross-functional teams verify dependencies and debug production systems.
- Refactored and modularized the liquid handler codebase to streamline error handling and sample preparation in NGS workflows, improving maintainability and accelerating lab operations across 20 Hamilton STARs.
- Developed barcode scanning systems and ETL workflows to integrate sample data into Sapio LIMS (laboratory information management systems) to automate sample tracking.

Lab Automation Engineer, **Sanofi Pasteur**

Oct 2022 – Nov 2024

Automation Team

- Wrote validation protocols and step-by-step instructions for 17 new users and trained them on the use of automated equipment.
- Optimized automated high-throughput methods on Hamilton platforms, reducing assay processing times by up to 30% across workflows including fluorescent-based assays and DNA extraction.
- Integrated devices such as Biotek, Cytomat, and plate readers with Hamilton's programming software by teaching plate to correct destination and importing device library to initialize software.
- Delivered biweekly presentations to key stakeholders, highlighting automation methods, technical challenges, implemented solutions, and supporting data that improved assay performance and usability.

Senior Research Associate, **Dermibiont Inc.**

Boston, MA

Skin Microbiome Startup

Oct 2021 – Sept 2022

- Built pipeline analysis for Oxford Nanopore Technology (ONT) using open source bioinformatic tools involving Guppy basecalling and quality control filtering.
- Collaborated with external bioinformatics teams to troubleshoot data pipelines and optimize QC logic for nanopore high-throughput sequencing workflows using Docker.
- Automated DNA extraction on Opentrons using Python for scripting and R for data visualization.
- Supported cloud-based data integration and analysis through Docker workflows and remote compute environments.

PUBLICATIONS

- Assavarittirong C, Au TY, Nguyen PV, Mostowska A. Vascular Ehlers-Danlos Syndrome: Pathological Variants, Recent Discoveries, and Theoretical Approaches. *Cardiol Rev.* 2022 Nov-Dec 01;30(6):308-313. doi: 10.1097/CRD.0000000000000419. Epub 2021 Sep 15. PMID: 34560710.
- Chudy, P., *et al.* (2024). Heme oxygenase-1 protects cells from replication stress. *Redox Biology*, 103247. ISSN 2213-2317. <https://doi.org/10.1016/j.redox.2024.103247>.

EDUCATION

Northeastern University

Boston, MA

Master's of Bioinformatics, concentration in Software Development

Sept 2026

Relevant Coursework: Python for Bioinformatics, Genomics in Bioinformatics, Algorithms, R programming

Uniwersytet Jagielloński

Krakow, PL

Master's in Molecular Biotechnology

Jul 2021

Master Thesis: Analysis of free heme levels and DNA replication progression in *Hmox1*-deficient HEK293T

University of California, Irvine

Irvine, CA

Bachelor of Science in Pharmaceutical Sciences

Sept 2019

Undergraduate Project: Characterization of human-derived 3'-5'-CyclicGMP aptamers